

QI Ecosystem — Component & Technology Inventory

Generated: 2026-08-02 · Owner: Renne Santiago / Quiddity Innovations · Maintainer: Claude Manager

This is the master inventory of every technology used across all QI projects, apps and products — what we created (proprietary QI code), what we borrowed (open-source components, with author, license, version in use, latest available version and repository link), what is vendored (copied in), and what was discussed but not adopted.

Data sources: per-app `status\_techstack.json` files in the project library (built 2026-06-24), a fresh scan of projects added since then (PlayDeck, Bakeoff, QI Connector, Headroom, video pipeline, MCP Gateway), and live PyPI / npm / GitHub version checks performed 2026-08-02.

1. Project index

Project / App	Where	Stack summary (top components)
AkiyaScout	C:\AkiyaScout	Python, FastAPI, Uvicorn, Pydantic, Gradio, selectolax
AutoPDF	C:\AutoPDF	Windows PowerShell, AutoPDF-Server.ps1, Vanilla JavaScript, HTML5 + CSS3, C#, Ghostscript
AvatarStudio	C:\AvatarStudio	Python, Gradio, Kokoro, edge-tts, soundfile + NumPy + SciPy, LivePortrait
Claude Manager	C:\CLAUDE	Claude Code, Opus / Sonnet / Haiku, Python, PowerShell, Windows Batch, Markdown front-matter agents
Claude Voice	C:\CLAUDE\Claude Voice	Python, FastAPI, Uvicorn, Pydantic, faster-whisper, edge-tts
CogniBase	C:\CogniBase	Python, FastAPI, Uvicorn, Pydantic, python-multipart, ChromaDB
CypherMiner	C:\CypherMiner	TypeScript, Vite, vite-plugin-pwa, Vanilla DOM, CSS variables, Web Crypto API
Digitization Cost Tool	(BU consulting tool)	Single-file HTML5, Vanilla JavaScript, Hand-written CSS, Web Storage, Browser print-to-PDF, Python 3 + python-docx
EasyFlow	C:\EasyFlow	Chrome / Edge Extension, JavaScript, Chrome Extension APIs, Google Apps Script, Python + Flask, Jinja2 + HTML/CSS/JS
FileHQ	C:\FileHQ	Python, FastAPI, Uvicorn, Jinja2 + StaticFiles, python-multipart, SQLite
Gamez	C:\Gamez	Single-file HTML/CSS/JS, Python, FastAPI, Uvicorn, httpx, APScheduler
LotteryWiz	C:\LotteryWiz	Python, FastAPI, Uvicorn, Python stdlib, NumPy, httpx
M2V	C:\M2V	Python, FastAPI, Uvicorn, python-multipart, Pydantic, Gradio
MQ	C:\MQ	Python, FastAPI, Uvicorn, Pydantic, Gradio, feedparser
MapSnap	C:\MapSnap	Python, Python stdlib http.server, Single-page HTML / CSS / vanilla JS, SQLite, JSON sidecar files, pyodbc
NEXUS	C:\NEXUS	Python, FastAPI, Uvicorn, Pydantic, Gradio, asyncio
Naya	C:\NAYA	Python, Flask, Anthropic SDK, requests, FastAPI + Uvicorn, Gradio
OpenClaw	C:\OC (WSL)	OpenClaw, WSL — Ubuntu, systemd, Python 3.11 / 3.12, Bash, Node.js
PersonalSong Studio	C:\PersonalSong	Python, FastAPI, Uvicorn, Pydantic, python-multipart + aiofiles, ACE-Step
QI Brain	C:\QIH\engine\brain	Python, FastAPI, Uvicorn, Pydantic, JSON-RPC 2.0 over stdio, SQLite
QI Hive	C:\QIH	Python, FastAPI, Uvicorn, Pydantic, AdminLTE, Bootstrap
TubeScout	C:\TUBESCOUT	Python, FastAPI, Uvicorn, Pydantic, Gradio, Server-rendered HTML + CSS + vanilla JS
PlayDeck *(new since 6/24)*	C:\PlayDeck, :8506	FastAPI, Uvicorn[standard], httpx, yt-dlp, curl_cffi, python-dotenv

Bakeoff *(new since 6/24)*	C:\QIP\Bakeoff	Hermes Agent, OpenClaw, Ollama, PyYAML, RDAP via stdlib urllib, WSL2
QI Connector *(new since 6/24)*	C:\QIP\Connector, :9030	MCP SDK / FastMCP, FastAPI, Uvicorn[standard], httpx, Pydantic, python-docx
Claude Manager Tools — video pipeline *(new since 6/24)*	C:\CLAUDE\Tools	edge-tts, Pillow, FFmpeg, python-docx, AWS CLI, winget + Task Scheduler
Headroom *(new since 6/24)*	C:\CLAUDE\Tools, :9020	headroom-ai[all], litellm, anthropic SDK, torch/transformers/huggingface-hub/datasets, opentelemetry api/sdk/otlp, NSSM
QI MCP Gateway *(new since 6/24)*	C:\QIH\engine\common\qi_mcp_gateway.py	MCP SDK / FastMCP + transport_security, httpx

2. Borrowed components — open-source registry

Everything below is third-party. In use = version pinned/observed in our projects; Latest = current upstream release as of the generation date.  = a newer upstream version exists ·  = we are current · blank = not comparable (unpinned/stdlib/service).

Languages & Runtimes

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
Python (CPython)	—	Apache 2.0; Apache 2.0 / Apache 2.0; BSD	0.0.20; 3.10+; 3.11	—		—	21 apps: AkiyaScout, AvatarStudio, Claude Manager, Claude Voice, CogniBase...
JavaScript (ES Modules / vanilla)	—	Built-in (ES2020); N/A (project source); Open standard	ES2022+; asset token v=20260624a; generated	—		—	7 apps: AutoPDF, CypherMiner, Digitization Cost Tool, EasyFlow, LotteryWiz...
Windows Batch (.bat)	—	Microsoft (OS); N/A; Windows	Windows 11; cmd.exe; n/a	—		—	AkiyaScout, Claude Manager, Digitization Cost Tool, QI Hive
PowerShell	—	BSD; MIT (PowerShell Core) / Windows; MIT (PowerShell) / proprietary (Windows PowerShell 5.1)	5.1 target, 7+ compatible; Windows PowerShell 5.1+; see r...	—		—	AutoPDF, Claude Manager, Claude Voice
Bash / Shell	—	GPL	5.x	—		—	OpenClaw
Google Apps Script (V8)	—	Free (Google quotas apply)	V8 runtime	—		—	EasyFlow
Node.js	—	MIT-style	system node	—		—	OpenClaw
OpenJDK / Java JRE	—	GPL-2.0-with-classpath-exception	bundled	—		—	AutoPDF
TypeScript	Microsoft	Apache-2.0	^5.6.3	7.0.2	 newer available	https://github.com/microsoft/TypeScript	CypherMiner
note	*major jump from pinned ^5.6.3 — TS7 native compiler; check migration notes*						

Web & API Frameworks / Servers

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
FastAPI	Sebastián Ramírez (tiangolo)	MIT	0.115+; 0.115.6; >=0.104	0.141.1	 newer available	https://github.com/fastapi/fastapi	15 apps: AkiyaScout, Claude Voice, CogniBase, CypherMiner, FileHQ...
note	*repo moved from tiangolo/fastapi to fastapi/fastapi org*						
Uvicorn	Encode (Tom Christie et al.)	BSD-3-Clause	0.30+; 0.34.0 [standard]; >=0.24 [standard]	0.52.1	 newer available	https://github.com/encode/uvicorn	14 apps: AkiyaScout, Claude Voice, CogniBase, CypherMiner, FileHQ...
Gradio	Gradio / Hugging Face	Apache-2.0	6.14.0; >=5.0	6.22.0	 newer available	https://github.com/gradio-app/gradio	7 apps: AkiyaScout, AvatarStudio, M2V, MQ, NEXUS...
note	*ahead of pinned 6.14.0*						
Jinja2 templating	Pallets Projects	BSD-3-Clause	3.1.5; Flask templating	3.1.6	 newer available	https://github.com/pallets/jinja	EasyFlow, FileHQ
Caddy	Matt Holt / Caddy team	Apache-2.0	engine/bin/caddy.exe	v2.11.4		https://github.com/caddyserver/caddy	QI Hive
Flask	Pallets Projects	BSD-3-Clause	3.x	3.1.3		https://github.com/pallets/flask	Naya

Frontend, UI & Styling

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
Single-file / single-page HTML-CSS-JS	—	N/A (project source); Proprietary (QI); Proprietary (Quiddity Innovations)	1 file; embedded <style>; n/a	—		—	AutoPDF, CypherMiner, Digitization Cost Tool, Gamez
Browser storage (localStorage / Web Storage)	—	Browser built-in; Web standard	with in-memory fallback; —	—		—	CypherMiner, Digitization Cost Tool
AdminLTE	ColorlibHQ (Colorlib)	MIT	4.x	v4.1.0		https://github.com/ColorlibHQ/AdminLTE	QI Hive
Bootstrap	Bootstrap core team (Mark Otto, Jacob Thornton et al.)	MIT	1.13.1; 5.x	v5.3.8	 newer available	https://github.com/twbs/bootstrap	QI Hive
folium / Leaflet	python-visualization team	MIT	>=0.20	0.20.0	 current	https://github.com/python-visualization/folium	AkiyaScout
note	*matches pinned floor*						
pywebview	Roman Sirokov (r0x0r)	BSD-3-Clause	launcher_gui.py	6.2.1		https://github.com/r0x0r/pywebview	Gamez
SortableJS	SortableJS org (RubaXa / community maintainers)	MIT	1.15.0	1.15.7	 newer available	https://github.com/SortableJS/Sortable	QI Hive
note	*ahead of pinned 1.15.0*						
Vite	Evan You / VoidZero,	MIT	^0.21.1; ^6.0.7	8.2.0	 newer available	https://github.com/vitejs/vite	CypherMiner

	Vite team						
note	*major jump from pinned ^6.0.7 — breaking changes likely*						

Data Validation, HTTP Clients & Concurrency

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
Pydantic	Samuel Colvin / Pydantic Services team	MIT	(FastAPI dep); 2.x; >=2.6	2.13.4	 newer available	https://github.com/pydantic/pydantic	11 apps: AkiyaScout, Claude Voice, CogniBase, CypherMiner, M2V...
httpx	Encode	BSD-3-Clause	>=0.25; >=0.27; current	0.28.1	 newer available	https://github.com/encode/httpx	10 apps: AkiyaScout, CogniBase, CypherMiner, Gamez, LotteryWiz...
requests	Python Software Foundation (originally Kenneth Reitz)	Apache-2.0	2.34.2; >=2.31; >=2.32	2.34.2	 current	https://github.com/psf/requests	7 apps: AvatarStudio, Claude Voice, FileHQ, M2V, MQ...
asyncio	—	MIT; PSF (stdlib)	>=8; stdlib	—		—	CogniBase, NEXUS, QI Brain
aiofiles	Tin Tvrtković	Apache-2.0	24.1.0; >=23.2	25.1.0	 newer available	https://github.com/Tinche/aiofiles	FileHQ, NEXUS
urllib (stdlib)	—	PSF (stdlib)	stdlib	—		—	LotteryWiz, MQ
humanize / rich	Will McGugan / Textualize	MIT	4.11.0 / 13.9.4	15.0.0	 newer available	https://github.com/Textualize/rich	FileHQ
note	*major bump from pinned 13.9.4*						
threading	—	PSF (stdlib)	stdlib	—		—	QI Brain

Databases, Vector Stores & On-Disk State

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
JSON files as state store	—	Internal (QI); MIT; MIT (spec)	-; 6.0.2; C:\QIH\ecosystem\	—		—	12 apps: AutoPDF, Claude Manager, Claude Voice, CogniBase, FileHQ...
SQLite	—	Public domain; Public domain / Apache 2.0	3.x; 3.x (WAL mode); 3.x (WAL mode, FK on)	—		—	12 apps: AkiyaScout, Claude Manager, CogniBase, FileHQ, LotteryWiz...
ChromaDB	Chroma (Jeff Huber, Anton Troynikov)	Apache-2.0	0.5.x (PersistentClient); >=0.5; local	1.5.9	 newer available	https://github.com/chroma-core/chroma	CogniBase, MapSnap, NEXUS, QI Brain, QI Hive
note	*major jump from pinned 0.5.x —						

	breaking changes likely*						
Excel COM as data mirror	—	Commercial (Microsoft)	Office 2013+	—		—	AutoPDF
oracledb (Oracle driver)	Oracle Corporation	Apache-2.0 / UPL (dual)	via /api/install-driver	4.0.2		https://github.com/oracle/python-oracledb	MapSnap
note	*PyPI name 'oracledb'*						
psycopg2 (PostgreSQL driver)	Federico Di Gregorio / psycopg team	LGPL-3.0 (linking exception)	via /api/install-driver	2.9.12		https://github.com/psycopg/psycopg2	MapSnap
PyMySQL (MySQL driver)	PyMySQL team	MIT	via /api/install-driver	1.2.0		https://github.com/PyMySQL/PyMySQL	MapSnap
pyodbc (SQL Server driver)	Michael Kleehammer	MIT	via /api/install-driver	5.3.0		https://github.com/mkleehammer/pyodbc	MapSnap
Qdrant	Qdrant team	Apache-2.0	optional	v1.18.3		https://github.com/qdrant/qdrant	MapSnap

AI — Local Inference (LLMs, Embeddings, Speech, Vision, Media)

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
Ollama	Ollama (org)	MIT	/api/tags, /api/chat; API :11434; HTTP (no SDK)	v0.32.5	✅ current	https://github.com/ollama/ollama	15 apps: AkiyaScout, AutoPDF, Claude Voice, CogniBase, EasyFlow...
note	*released 2026-07-27; local server, not a pip package*						
faster-whisper / Whisper (STT)	SYSTRAN (Guillaume Klein)	MIT	>=20231117; base model; model base.en (PC) / base (multil...	1.2.1		https://github.com/SYSTRAN/faster-whisper	Claude Voice, M2V, OpenClaw, PersonalSong Studio, TubeScout
edge-tts (Microsoft neural voices)	rany2	LGPL-3.0 (srt_composer.py MIT)	7.2.8; Communicate API; dots.tts 2B	7.2.8	✅ current	https://github.com/rany2/edge-tts	AvatarStudio, Claude Voice, OpenClaw, PersonalSong Studio
note	*matches version we run*						
Local open-weight LLMs (Qwen/Gemma/DeepSeek/etc.)	—	Derived from base model (e.g. Qwen2.5); Free tier / local; Mixed open weights / MIT runtime	autopdf-mapper:latest; see NEXUS providers.json; via Ollama	—		—	AutoPDF, Naya, TubeScout
PyTorch (CUDA)	PyTorch Foundation / Meta	BSD-3-Clause	2.11.0+cu128; >=2.0 (torch 2.11 / cu128 build); torch 2.1...	2.13.0	📌 newer available	https://github.com/pytorch/pytorch	AvatarStudio, M2V, PersonalSong Studio
note	*we run 2.11.0+cu128*						
Kokoro-82M (offline TTS)	hexgrad	Apache-2.0	0.9.4; planned	unverified		https://huggingface.co/hexgrad/Kokoro-82M	AvatarStudio, Claude Voice
note	*HF model repo, no release tags; last						

	modified 2025-04-10*						
librosa / audio analysis	librosa development team (Brian McFee et al.)	ISC	>=0.10; soundfile (libsndfile) · pydub · scipy>=1.11 · nu...	0.11.0	✅ current	https://github.com/librosa/librosa	M2V, PersonalSong Studio
Stable Diffusion XL / SVD (diffusers)	Hugging Face	Apache-2.0	diffusers>=0.32 / transformers>=4.44 / accelerate>=0.33; ...	0.39.0	✅ current	https://github.com/huggingface/diffusers	M2V, PersonalSong Studio
ACE-Step (singing model)	ACE Studio / StepFun	Apache-2.0	0.2.0 (per song metadata)	unverified		https://github.com/ace-step/ACE-Step	PersonalSong Studio
note	*no GitHub releases; we run 0.2.0 locally*						
ComfyUI	Comfy-Org (formerly comfyanonymous)	GPL-3.0	core nodes	v0.29.2		https://github.com/Comfy-Org/ComfyUI	M2V
note	*old comfyanonymous URL redirects here*						
Demucs (stem separation)	Alexandre Défossez (orig. Meta/FAIR, facebookresearch repo archived)	MIT	htdemucs, two-stems mode	4.1.0		https://github.com/adefossez/demucs	PersonalSong Studio
note	*adefossez fork is the maintained one; PyPI points to it*						
IP-Adapter (character consistency)	Tencent AI Lab	Apache-2.0	ip-adapter_sd-xl	unverified		https://github.com/tencent-ailab/IP-Adapter	M2V
note	*weights at huggingface.co/h94/IP-Adapter ; no releases*						
NotebookLM (vision-driven)	—	Commercial (free)	—	—		—	OpenClaw
OpenCV (computer vision)	OpenCV.org / opencv-python maintainers	Apache-2.0	opencv-python (cv2)	5.0.0.93		https://github.com/opencv/opencv-python	AvatarStudio
rembg (background removal)	Daniel Gatis	MIT	rembg 2.0.75 / pillow 12.2.0	2.0.77	📌 newer available	https://github.com/danielgatis/rembg	AvatarStudio
note	*we run 2.0.75*						
Seed-VC (voice conversion)	Plachtaa	GPL-3.0	f0-conditioned (sing) / non-f0 (speech)	unverified		https://github.com/Plachtaa/seed-vc	PersonalSong Studio
note	*no releases; latest commit 2025-04-20*						
Talking-head render engines (Hallo2/LivePortrait/SadTalker/etc.)	—	Apache 2.0 (research); MIT (research); Mixed (research)	WSL2 conda env 'hallo'; WSL2 conda env 'liveportrait'; WS...	—		—	AvatarStudio
VAD & wake-word (silero/web rtc/openWakeWord)	Silero Team	MIT	see requirements.txt	6.2.1		https://github.com/snakers4/silero-vad	Claude Voice

AI — Cloud & Hosted Models

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
Anthropic Claude (API /	Anthropic	MIT	OpenAI-compatible /	0.120.2	📌 newer available	https://github.com/anthropics/anthropic-	9 apps: Claude

SDK)			vendor APIs; Opus 4.x / Sonnet 4.x / ...			sdk-python	Manager, Claude Voice, CogniBase, EasyFlow, Gamez...
OpenAI / ChatGPT (API)	OpenAI	Apache-2.0	gpt-4o-mini; gpt-5.4; openai>=1.50 / google-generativeai>...	2.52.0	✅ current	https://github.com/openai/openai-python	CogniBase, EasyFlow, NEXUS, OpenClaw
Cloudflare Workers AI	—	Commercial (free tier); Free tier	llama-3.3-70b-fp8; qwen2.5-7b-instruct-fast; —	—		—	MQ, NEXUS, OpenClaw
OpenRouter (model gateway)	—	Commercial API; Free tier; Mixed (mostly paid)	/api/v1; see llm_chain; see providers.json	—		—	MapSnap, NEXUS, Naya
D-ID / HeyGen (avatar SaaS)	—	Commercial SaaS	D-ID v1, HeyGen v2	—		—	AvatarStudio
DeepEval (LLM evaluation)	Confident AI (Jeffrey Ip)	Apache-2.0	optional dep	4.1.5		https://github.com/confident-ai/deepeval	NEXUS
DeepL (translation)	—	Paid / free-tier API	v2	—		—	AkiyaScout
Google Gemini	—	Free tier / Paid	gemini-2.0-flash	—		—	EasyFlow
Groq	—	Free tier	llama-3.3-70b	—		—	NEXUS
Mistral	—	Free tier	mistral-small-latest	—		—	NEXUS
tiktoken	OpenAI (Shantanu Jain)	MIT	>=0.7	0.13.0	📌 newer available	https://github.com/openai/tiktoken	CogniBase

Document, PDF, Office & Media Processing

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
FFmpeg	FFmpeg community project	LGPL-2.1+/GPL-2.0+ (dual, build-dependent)	Pillow>=10 / imageio>=2.34; WSL2 system ffmpeg + imageio-...	8.1.2 ('Hoare')	📌 newer available	https://github.com/FFmpeg/FFmpeg	AvatarStudio, Claude Voice, M2V, PersonalSong Studio
note	*canonical repo git.ffmpeg.org ; GitHub is a mirror*						
lxml / selectolax / BeautifulSoup (HTML/XML parsing)	lxml dev team (Stefan Behnel et al.)	BSD-3-Clause	>=0.3.21; >=5.1; current	6.1.1	📌 newer available	https://github.com/lxml/lxml	AkiyaScout, CogniBase, LotteryWiz
openpyxl	openpyxl team (Charlie Clark, maintainer)	MIT	>=3.1; current; installed	3.1.5	📌 newer available	https://foss.heptapod.net/openpyxl/openpyxl	CogniBase, LotteryWiz, MapSnap
note	*canonical repo is Heptapod (Mercurial), not GitHub*						
PDF text extraction (PyMuPDF/Poppler/PyPDF2/pdfminer)	Artifex Software Inc.	AGPL-3.0 OR Artifex commercial (dual)	>=1.24; bundled; optional	1.28.0	📌 newer available	https://github.com/pymupdf/PyMuPDF	AutoPDF, CogniBase, MapSnap
Microsoft Office COM / pywin32	Mark Hammond et	PSF-2.0	Office 2013+; optional	312		https://github.com/mhammond/pywin32	AutoPDF, Naya

	al.						
note	*build-number versioning*						
mutagen (audio tags)	Quod Libet project (Christoph Reiter et al.)	GPL-2.0+	>=1.47; soundfile>=0.12 / mutagen>=1.47	1.48.1	 newer available	https://github.com/quodlibet/mutagen	M2V, PersonalSong Studio
soundfile / pydub / scipy / numpy (audio I/O)	Bastian Bechtold	BSD-3-Clause	see requirements.txt; soundfile 0.13.1 / numpy 2.4.5 / sc...	0.14.0	 newer available	https://github.com/bastibe/python-soundfile	AvatarStudio, Claude Voice
note	*newer than our 0.13.1*						
Ghostscript	Artifex Software Inc.	AGPL-3.0 OR Artifex commercial (dual)	10.07.0	10.07.1	 newer available	https://github.com/ArtifexSoftware/ghostpdl	AutoPDF
note	*we run 10.07.0 — one patch behind*						
matplotlib + pandas	John D. Hunter / matplotlib dev team	Matplotlib License (PSF/BSD-style)	>=3.8 / >=2.2	3.11.1	 newer available	https://github.com/matplotlib/matplotlib	CogniBase
MoviePy	Zulko et al.	MIT	v2 API (with_duration/with_effects)	2.2.1		https://github.com/Zulko/moviepy	M2V
NAPS2 (scanner bridge)	Ben Olden-Cooligan (cyanfish)	GPL-2.0	8.2.1	8.3.2	 newer available	https://github.com/cyanfish/naps2	AutoPDF
note	*one minor ahead of our 8.2.1*						
NumPy / SciPy	NumPy Developers / NumFOCUS	BSD-3-Clause	optional	2.5.1		https://github.com/numpy/numpy	LotteryWiz
PDFtk (pdftk-java)	Angad Singh (maintainer); originally Sid Steward's pdftk	GPL-2.0+	bundled	3.3.3		https://gitlab.com/pdftk-java/pdftk	AutoPDF
note	*canonical is GitLab*						
Pillow (PIL)	Jeffrey A. Clark and contributors	HPND ('Pillow License')	bundled	12.3.0		https://github.com/python-pillow/Pillow	PersonalSong Studio
note	*newer than our 12.2.0*						
qrcode	Lincoln Loop	BSD-3-Clause	bundled	8.2		https://github.com/lincolinloop/python-qrcode	PersonalSong Studio
RapidOCR (ONNX)	SWHL / RapidAI team	Apache-2.0	not started (plan 2026-06- 22)	1.4.4		https://github.com/RapidAI/RapidOCR	AutoPDF
note	*unified 'rapidocr' package now also exists*						
Tabula (tabula-java)	Manuel Arístarán / Tabula project	MIT	bundled (optional)	1.0.5		https://github.com/tabulapdf/tabula-java	AutoPDF
note	*last release 2021 — dormant*						
Tesseract OCR	HP-originated;	Apache-2.0	bundled	5.5.3		https://github.com/tesseract-ocr/tesseract	AutoPDF

	Google + community (lead Zdenko Podobny)						
ytmusicapi	sigma67	MIT	>=1.10	1.12.1	 newer available	https://github.com/sigma67/ytmusicapi	PersonalSong Studio
zipfile + xml (stdlib)	—	PSF	stdlib	—		—	MapSnap

Service Management, Tunnels & Scheduling

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
NSSM (Windows service manager)	Iain Patterson	Public Domain	2.24 (engine/bin/nssm.exe); C:\QIH\engine\bin\nssm.exe; n...	2.24	 current	https://nssm.cc	18 apps: AutoPDF, AvatarStudio, Claude Manager, Claude Voice, CogniBase...
note	*canonical source is SVN at nssm.cc, not GitHub; 2.24 (2014) still latest — project dormant*						
APScheduler	Alex Grönholm	MIT	3.10.4; >=3.10	3.11.3	 newer available	https://github.com/agronholm/apscheduler	AkiyaScout, FileHQ, Gamez, NEXUS, Naya
Cloudflare Tunnel (cloudflared)	Cloudflare, Inc.	Apache-2.0	named tunnel qi-kaze; named tunnels (migrated 2026-06-20)...	2026.7.3		https://github.com/cloudflare/cloudflared	MapSnap, OpenClaw, QI Hive
Windows Task Scheduler	—	OS; OS built-in; OS component	Windows 11; —	—		—	MQ, Naya, TubeScout
WSL / WSL2 (Ubuntu)	—	Open source; Ubuntu (Canonical) / WSL (MS)	Ubuntu-24.04	—		—	AvatarStudio, OpenClaw
PyInstaller	PyInstaller Development Team	GPL-2.0 w/ bundling exception	WC2026.spec	6.21.0		https://github.com/pyinstaller/pyinstaller	Gamez
setuptools + Docker	—	MIT / Apache 2.0	pyproject 0.4.0.dev0	—		—	CogniBase
Win32 process control (CIM/Stop-Process)	—	Windows	CIM	—		—	Claude Manager

External Integrations & Data Sources

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
feedparser (RSS/Atom)	Kurt McKee	BSD-2-Clause	>=6.0.11; —	6.0.14	 newer available	https://github.com/kurtmckee/feedparser	MQ, NEXUS, OpenClaw
LINE Messaging API	LINE Corporation	Apache-2.0	messaging v3; v2 broadcast; —	3.25.0		https://github.com/line/line-bot-sdk-python	Claude Manager, Claude Voice, OpenClaw
note	*messaging v3 SDK*						
Gmail API / IMAP-SMTP	—	Commercial (free); Free (Google quotas apply)	v1; —	—		—	EasyFlow, OpenClaw
Google Workspace APIs	Google LLC	Apache-2.0	>=1.2 / >=0.2; v1	2.198.0	 newer available	https://github.com/googleapis/google-	EasyFlow, TubeScout

(Apps Script / People / Drive)						api-python-client	
Chrome / Edge Extension platform (Manifest V3)	—	Chrome extension APIs; Open standard; Open standard (Google)	10 locales; MV3; manifest v1.2.1	—		—	EasyFlow
Facebook / Instagram / WhatsApp (Meta APIs)	—	Meta Platform Terms	planned; v19.0	—		—	MQ
Lottery open-data feeds	—	Open data / official	live	—		—	LotteryWiz
Map tiles & geocoding (OSM/Esri/GSI)	—	ODbL (free, usage policy); Tile-provider terms (attribution required)	live tiles; public API	—		—	AkiyaScout
Plex (media server)	—	in-repo module	local Plex Media Server :32400	—		—	PersonalSong Studio
praw (Reddit)	Bryce Boe / PRAW dev team	BSD-2-Clause	>=7.8	8.0.2	↑ newer available	https://github.com/praw-dev/praw	NEXUS
Sports & odds data APIs (ESPN/API-Football/Kalshi/EA FC)	—	Community dataset; Free tier (paid plans); Public API (no key)	GitHub EAFC26-DataHub; site.api.espn.com v2/v3; trade-api/v2	—		—	Gamez
Telegram Bot API	—	Free API	Bot API	—		—	Naya
YouTube Data API & transcripts	Jonas Depoix	MIT	>=0.6.2	1.2.4	↑ newer available	https://github.com/jdepoix/youtube-transcript-api	TubeScout
yt-dlp	yt-dlp maintainers collective (fork of youtube-dl)	Unlicense (public domain)	module -m yt_dlp	2026.7.4		https://github.com/yt-dlp/yt-dlp	PersonalSong Studio

Security, Auth & Cryptography

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
stdlib crypto (hmac / hashlib / secrets)	—	PSF; PSF (stdlib)	stdlib	—		—	MapSnap, Naya
Authlib	Hsiaoming Yang (lepture)	BSD-3-Clause	>=1.3	1.7.2	↑ newer available	https://github.com/authlib/authlib	CogniBase
Google OAuth2 (+ PKCE)	—	Open standard	OAuth 2.0	—		—	EasyFlow
Hashing for dedup (xxhash / hashlib)	Yue Du	BSD-2-Clause	3.5.0	3.8.1	↑ newer available	https://github.com/ifduyue/python-xxhash	FileHQ
note	*newer than our 3.5.0*						
sqlglot (SQL safety)	Toby Mao	MIT	>=25.0	30.14.0	↑ newer available	https://github.com/tobymao/sqlglot	CogniBase
Web Crypto API	—	Browser built-in	—	—		—	CypherMiner

Developer Tooling, Agents & Protocols

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
Git / GitHub	—	GPL; GPL /	C:\Program	—		—	Claude Manager,

		proprietary; GPLv2	Files\Git\cmd\git.exe; git; system git				OpenClaw, QI Brain, QI Hive
Claude Code (Anthropic CLI / Agent SDK)	—	Commercial (Anthropic); Internal (QI); Subscription (Anthropic)	Claude Code / Desktop (MSIX); engine/common/usage_stats.p...	—		—	Claude Manager, Claude Voice, QI Hive
Model Context Protocol (MCP / FastMCP)	Jeremiah Lowin	Apache-2.0	API on port 9011; JSON-RPC 2.0; mcp.server.fastmcp	3.4.5	↑ newer available	https://github.com/jlowin/fastmcp	Claude Manager, LotteryWiz, QI Hive
note	*homepage gofastmcp.com*						
rapidfuzz / difflib (fuzzy matching)	Max Bachmann	MIT	>=3.9; stdlib	3.14.5	↑ newer available	https://github.com/rapidfuzz/RapidFuzz	NEXUS, Naya
filesystem stdlib (os/shutil/pathlib)	—	PSF (stdlib)	stdlib	—		—	FileHQ
langdetect	Michal 'Mimino' Danilák	Apache-2.0	optional	1.0.9		https://github.com/Mimino666/langdetect	Claude Voice
note	*Python port of Google's language-detection*						
Markdown front-matter agents	—	n/a (project files)	8 agents	—		—	Claude Manager
Playwright / Pester (UI & API tests)	Microsoft Corporation	Apache-2.0	Tests/ tree	1.62.0		https://github.com/microsoft/playwright-python	AutoPDF
send2trash	Andrew Senetar (arsenetar)	BSD-3-Clause	>=1.8	2.1.0	↑ newer available	https://github.com/arsenetar/send2trash	Naya

QI Internal Platform Services

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
NEXUS (multi-AI synthesis)	—	QI internal; Quiddity Innovations (internal)	0.1.0; registry; —	—		—	FileHQ, Naya, TubeScout
Maia / Naya shared modules	—	QI internal; Quiddity Innovations (internal)	merged 2026; shared with Maia	—		—	FileHQ, Naya
OpenClaw (agent gateway)	—	Open source (npm package)	2026.4.26 (be8c246)	—		—	OpenClaw

Other / Project-Specific Components

Component	Author / Maintainer	License	In use	Latest	Status	Repository	Used in
Browser print-to-PDF	—	Browser-native	@media print	—		—	Digitization Cost Tool
Built-in glossary stub	—	In-house	n/a	—		—	AkiyaScout
file:/// · web server · SharePoint embed	—	N/A	no install	—		—	Digitization Cost Tool
OnBase XML Index DIP	—	n/a (output format)	Foundation 24.1	—		—	AutoPDF

smtplib (stdlib)	—	PSF (stdlib)	stdlib	—		—	AkiyaScout
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Additional components from the post-6/24 projects

Component	Author / Maintainer	License	Latest	Repository	Used in
curl_cffi	lexiforest (Yifei Kong)	MIT	0.16.0	https://github.com/lexiforest/curl_cffi	PlayDeck
python-dotenv	Saurabh Kumar (theskumar)	BSD-3-Clause	1.2.2	https://github.com/theskumar/python-dotenv	PlayDeck
PyYAML	YAML community (orig. Kirill Simonov)	MIT	6.0.3	https://github.com/yaml/pyyaml	Bakeoff
litellm	BerriAI (Ishaan Jaffer, Krrish Dholakia)	MIT	1.95.0	https://github.com/BerriAI/litellm	Headroom
headroom-ai	Headroom project	Apache-2.0	0.33.0	https://headroom-docs.vercel.app	Headroom (QI_Headroom :9020)
hls.js	video-dev community	Apache-2.0	1.6.16	https://github.com/video-dev/hls.js	PlayDeck (vendored)

3. QI-created components (proprietary)

Original Quiddity Innovations code — no upstream, no external author.

Component	What it is	Lives in
Single-file HTML/CSS/JS front-ends	Hand-written no-framework UIs (MapSnap schema browser, dashboards, launchers)	MapSnap, AutoPDF, Naya, many
QI Hive dashboard + War Room + Library	FastAPI ops center: mission control, dispatch, compliance, doc Library (:8600)	C:\QIH\engine\hive\dashboard
QI Brain (API + ChromaDB memory)	Ecosystem memory: decisions, sessions, features, doc index (:9011)	C:\QIH\engine\brain
qi_mcp_gateway.py	Reusable config-driven MCP front door for any QI app (ADAPTERS registry); MapSnap first adopter	C:\QIH\engine\common
QI Connector	Remote MCP server for claude.ai / Claude Code (:9030, connector.quiddityinnovations.com)	C:\QIP\Connector
QI Elevate broker	Elevation broker for headless admin ops (replaces gsudo)	C:\QIH\engine\common
qi_registry.py / qi_validator.py / qi_new_project.py	Ecosystem registry, compliance checker, project scaffolder	C:\QIH\ecosystem
Maia / Naya / NEXUS bot engines	Multi-channel AI assistant platform, multi-LLM chains, template engine vision	C:\QI, C:\NAYA, C:\NEXUS
MapSnap engine + extractors	DB schema browser/extractor suite (SQL Server/Postgres/MySQL/Oracle/Qdrant), Data Chat, BU Edition kit	C:\MapSnap
AutoPDF engine	PowerShell HttpListener PDF/OCR pipeline	C:\AutoPDF
EasyFlow extension	Chrome/Edge MV3 extension (10 locales)	C:\EasyFlow
qi_toon_video.py + BU video pipeline	Flat-2D character animation / explainer-video generator (edge-tts + Pillow + FFmpeg)	C:\CLAUDE\Tools
doc_harvester.py + hive-librarian	Documentation Brain: doc index + knowledge graph (937+ docs)	C:\QIH\engine\brain
Bakeoff eval rig	Hermes vs OpenClaw agent-harness evaluation (same gpt-oss-20b brain)	C:\QIP\Bakeoff
PlayDeck	Own video player UI: 3-tier browse (yt-dlp/scrape/headless), offline library (:8506)	C:\PlayDeck
PersonalSong Studio	Local AI song generator orchestration (ACE-Step + Demucs + Seed-VC)	C:\PersonalSong

TubeScout	YouTube-subscription news tool (:8503)	C:\TUBESCOUT
Claude Voice	LINE/Telegram voice bridge + meeting room (:8722)	C:\CLAUDE\Claude Voice
Hive agent suite	8 specialized sub-agents (architect/builder/inspector/ops/scout/scribe/tester/librarian) + dispatch protocol	C:\CLAUDE\claude\agents

4. Vended third-party code (copied in, not package-managed)

- C:\PlayDeck\web\vendor\hls.min.js — vendored minified hls.js build (no version string)
- C:\PlayDeck\engine\bin\nssm.exe — per-project NSSM copy (QI convention)
- C:\QIP\Connector — NSSM per-project service convention (shared C:\QIH\engine\bin\nssm.exe)
- C:\CLAUDE\Tools\headroom_env\ — entire isolated Python venv vendored (torch, transformers, datasets, cv2, aiohttp, cryptography...), no manifest
- No verbatim code copying between QI sibling projects found — cross-project reuse is the shared nssm.exe binary and the intentionally shared qi_mcp_gateway.py module
- `C:\QIH\engine\hive\dashboard\static\vendor\vis-network.min.js` — vis-network graph library (Apache-2.0/MIT dual), used by the Library knowledge graph
- Per-project `engine\bin\nssm.exe` copies across QI projects (standing convention: each project ships its own NSSM 2.24)
- Bundled Java runtimes + tabula-java / pdftk-java jars inside AutoPDF (no system Java install required)

5. Components discussed but NOT adopted (or only partially adopted)

- Buzz (Block / Jack Dorsey's open-source AI-agent workspace) — evaluated as successor to the hand-built QI Hive War Room/dispatch UI; not migrated (too new; replatforming mid-cycle would burn August). Sandbox trial + 3–6 month re-check instead. (2026-07-27, C:\CLAUDE\Tech_Evaluation_2026-07.md)
- Colibri (MoE model streaming engine, 744B GLM-5.2 on consumer hardware) — evaluated for "Ronald" local Ollama fallback; watch-list only — ~9 tok/s too slow for voice latency. (2026-07-27, Tech_Evaluation_2026-07.md)
- Nostr (decentralized signed-message protocol) — evaluated as notification/identity layer; skipped — LINE/Telegram already cover it. (2026-07-27, Tech_Evaluation_2026-07.md)
- OpenClaw as the Claude Voice harness — rejected for the real-time voice/video rebuild (gateway 404/version mismatch, fragile extra hop); LiveKit chosen instead. (2026-06-21, ClaudeVoice summaries)
- MiniMax 230B local — rejected as MapSnap local LLM: 108 GB at 4-bit, needs 128 GB RAM — not viable on RTX 5080/64 GB. Cloud A/B variant also deferred. (2026-06-16/17, MapSnap summaries)
- Tailscale / Cloudflare Tunnel / raw port exposure for MapSnap BU Edition — rejected; IIS + VPN chosen for the ~100-user BU server (authorization stays in MapSnap's own login; no per-seat cost). (Brain decision_502, 2026-07)
- Headless Claude Code as Hive Auto-Apply executor — deferred then replaced by a deterministic Python worker (Dependabot/Renovate pattern) for reliability/auditability. (2026-05-13/14, Auto_Apply_Pipeline_Design)
- gsudo for headless elevation — superseded by the QI_Elevate broker; Access Denied from non-interactive subprocesses. Still fine interactively. (2026-05-14)
- RapidOCR / RapidOcrNet (PP-OCRv5/v6 ONNX+.NET) — approved 2026-06-22 as AutoPDF's second OCR engine, NOT started; Tesseract remains sole live engine. (AutoPDF status files)
- PaddlePaddle Python OCR prototype (autopdf_ocr.py) — rejected for footprint/AVX risk; never wired in; superseded by the planned RapidOcrNet route. (2026-06-22)
- Qdrant — QdrantExtractor built into MapSnap's extraction layer but shipped optional, never enabled by default. (MapSnap status files)
- ScanSnap scanners — rejected for Digitization-Costs tool (no TWAIN/ISIS); Ricoh fi-8170 chosen. (2026-06-12/15)

- pythonw.exe for scheduled tasks — rejected (nulls sys.stdout, breaks logging; venvs lack it); conhost --headless + python.exe instead. (2026-06-18)
- Cloudflare Workers AI via OpenClaw plugin — plugin found hardcoded to Anthropic API; patch deferred. (2026-05-16)
- dots.tts (local GPU voice cloning for OpenClaw Koe) — built but parked/disabled (EN/ZH only); edge-tts + Piper used instead. (OpenClaw status intro)
- D-ID / HeyGen (cloud avatar SaaS) — wired into AvatarStudio as swappable engines but disabled by default (API keys, cost); local LivePortrait/Hallo2 + Kokoro/edge-tts is the default path.
- MuseTalk / SadTalker / Wav2Lip — pre-wired AvatarStudio lip-sync alternates, disabled pending WSL2 installs (install-blocked more than rejected).
- XTTS-v2 / F5-TTS voice cloning — reserved as Stage 2.5 polish for War Room avatars; edge-tts is the live baseline, Kokoro the local middle ground. (2026-06-18, Phase N spec)
- WebSocket push for Naya — polling refresh used instead. (2026-04-13)
- Microsoft Graph API (Outlook) — deferred; Outlook→Gmail forwarding as interim. (2026-04-14)
- Playwright sync_testers.py for EasyFlow — deferred by Renne in favor of Claude-driven manual syncing. (2026-04-15)
- Cloudflare named tunnels for the 4-bot rollout — stayed on quick tunnels until the domain was owned (later adopted ecosystem-wide — partial/timing deferral). (2026-05-15)
- Telethon janitor (Telegram cleanup) — deferred; wire filter judged sufficient. (2026-05-15)
- webcall_server.py (WebRTC-style call server, Claude Voice) — superseded by meeting_server (both :8722); not run.

False lead checked: NotebookLM IS adopted (Kaze news-digest archive + knowledge-base client) — not a rejected item.

Appendix — how to regenerate this inventory

1. Rebuild per-app stacks: re-run the project-library build (`C:\QIH\shared\documentation\project\_library\\_build\_project\_docs.py`) so each app's `source/status\_techstack.json` is fresh.
2. Re-aggregate: `python C:\CLAUDE\Tools\techstack\_aggregate.py`
3. Refresh latest-version research (PyPI/npm/GitHub) and update the research JSONs.
4. `python C:\CLAUDE\Tools\gen\_component\_inventory.py` → writes this file into the project library.
5. `python C:\CLAUDE\Tools\gen\_inventory\_pdf.py` → refreshes the .docx/.pdf twins.
6. Re-index the Library: `python C:\QIH\engine\brain\doc\_harvester.py --no-embed` (or full run to embed).